

UNIT IV – MULTIMODAL GENERATIVE AI APPLICATIONS

Question 1: AI Chatbot for Engineering College Student Queries

Practical Implementation Report

Develop an AI chatbot that answers student queries related to an engineering college using a pre-trained language model.

1. Aim

To develop an AI chatbot that accepts engineering-college-related student queries and generates concise responses using a pre-trained language model.

2. Objective

- To provide a simple conversational interface for students.
- To use a pre-trained language model without training a model from scratch.
- To accept natural-language questions related to college courses, departments, examinations, library, laboratories, and campus facilities.
- To display the generated response through a Streamlit web application.

3. Software and Hardware Requirements

Software Requirements

- Python 3.x
- Visual Studio Code
- Streamlit
- Hugging Face Transformers
- PyTorch
- Hugging Face pre-trained model: google/flan-t5-small
- Windows operating system with internet access for the initial model download.

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM recommended
- Internet connection for downloading the pre-trained model and installing packages

4. Folder Structure

```
UNIT_IV_Multimodal_Generative_AI
├── Q1_College_Chatbot
│   ├── chatbot.py
│   └── venv
```

5. Libraries Used

- Streamlit – creates the interactive web interface.
- Transformers – loads and uses the pre-trained language model.
- PyTorch – provides the deep-learning model execution environment.
- AutoTokenizer – converts input text into model-readable tokens.
- AutoModelForSeq2SeqLM – loads the sequence-to-sequence language model.

6. Procedure / Step-by-Step Implementation

1. Create the main folder UNIT_IV_Multimodal_Generative_AI.
2. Create the Q1_College_Chatbot folder inside the main folder.
3. Create chatbot.py inside the Q1 folder.
4. Create and activate a Python virtual environment.

5. Install Streamlit, Transformers, PyTorch, and SentencePiece.
6. Import the required Python libraries.
7. Load the pre-trained google/flan-t5-small model and tokenizer.
8. Create a Streamlit interface with a question input box and Get Answer button.
9. Accept a student's engineering-college-related question.
10. Tokenize the question and send it to the pre-trained model.
11. Generate the response and decode it into readable text.
12. Display the generated answer in the Streamlit application.
13. Run the application using streamlit run chatbot.py and test multiple questions.

7. Implementation Code

```
import streamlit as st
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

st.set_page_config(
    page_title="Engineering College AI Chatbot",
    page_icon="🎓",
    layout="centered"
)

st.title("🎓 Engineering College AI Chatbot")

st.write(
    "Ask questions related to engineering college "
    "courses, departments, examinations, library, "
    "laboratories, and campus facilities."
)

@st.cache_resource
def load_model():
    model_name = "google/flan-t5-small"

    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

    return tokenizer, model

with st.spinner("Loading AI model..."):
    tokenizer, model = load_model()

question = st.text_area(
    "Enter your question:",
    placeholder=(
        "Example: What facilities are available "
        "in the Computer Science department?"
    )
)
```

```

)

if st.button("🤖 Get Answer"):

    if question.strip() == "":
        st.warning("Please enter a question.")

    else:
        prompt = f"""
You are an AI assistant for an engineering college.

Answer the student's question clearly and briefly.

Student Question:
{question}

Answer:
"""

        inputs = tokenizer(
            prompt,
            return_tensors="pt",
            truncation=True,
            max_length=512
        )

        with st.spinner("Generating answer..."):
            outputs = model.generate(
                **inputs,
                max_new_tokens=120,
                num_beams=4,
                early_stopping=True
            )

            answer = tokenizer.decode(
                outputs[0],
                skip_special_tokens=True
            )

            st.subheader("🤖 AI Answer")
            st.success(answer)

st.divider()

st.caption(
    "Powered by a pre-trained language model "
    "using Hugging Face Transformers."
)

```

8. Execution Command

```
streamlit run chatbot.py
```

9. Input and Output

Input 1:

What facilities are available in the computer science department?

Observed Output:

computer science labs

Input 2:

What is an engineering college?

Observed Output:

AI assistant

Input 3:

What facilities are available in the library?

Observed Output:

library

Input 4:

What is the purpose of laboratory sessions?

Observed Output:

To help students learn how to use a computer.

Input 5:

What are the common activities conducted by engineering students?

Observed Output:

Computer science and engineering

10. Screenshots of the Implemented Application

The following screenshots document successful execution of the Streamlit chatbot with different student queries.



Engineering College AI Chatbot

Ask questions related to engineering college courses, departments, examinations, library, laboratories, and campus facilities.

Enter your question:

What facilities are available in the computer science department?

Get Answer

AI Answer

computer science labs

Powered by a pre-trained language model using Hugging Face Transformers.

Figure 1. Engineering College AI Chatbot – Test Case 1



Engineering College AI Chatbot

Ask questions related to engineering college courses, departments, examinations, library, laboratories, and campus facilities.

Enter your question:

What is an engineering college?

Get Answer

AI Answer

AI assistant

Figure 2. Engineering College AI Chatbot – Test Case 2



Engineering College AI Chatbot

Ask questions related to engineering college courses, departments, examinations, library, laboratories, and campus facilities.

Enter your question:

What facilities are available in the library?

🗨️ Get Answer

😊 AI Answer

library

Powered by a pre-trained language model using Hugging Face Transformers.

Figure 3. Engineering College AI Chatbot – Test Case 3



Engineering College AI Chatbot

Ask questions related to engineering college courses, departments, examinations, library, laboratories and campus facilities.

Enter your question:

What is the purpose of laboratory sessions?

🗨️ Get Answer

🗨️ AI Answer

To help students learn how to use a computer.

Powered by a pre-trained language model using Hugging Face Transformers.

Figure 4. Engineering College AI Chatbot – Test Case 4



Engineering College AI Chatbot

Ask questions related to engineering college courses, departments, examinations, library, laboratories, and campus facilities.

Enter your question:

What are the common activities conducted by engineering students?

🗨️ Get Answer

🗨️ AI Answer

Computer science and engineering

11. System Workflow

The chatbot follows the workflow: Student Query → Tokenization → Pre-trained Language Model → Text Generation → AI Answer displayed in Streamlit.

12. Graph / Chart Requirement

A numerical graph is not required by the question. The question asks for development of an AI chatbot, so the most relevant visual evidence is the application interface and test-case screenshots. Therefore, screenshots are included instead of an artificial performance graph.

13. Result

The Engineering College AI Chatbot was successfully developed using Python, Streamlit, Hugging Face Transformers, and the pre-trained google/flan-t5-small language model. The application accepts student queries through a web interface and generates responses using the pre-trained model.

14. Conclusion

The developed chatbot demonstrates the use of a pre-trained language model for a practical educational application. It provides a simple interface through which engineering students can submit natural-language queries and receive automatically generated responses. The implementation can be extended with a college-specific knowledge base to improve factual accuracy and answer quality.

15. Viva / Demonstration Points

- The application uses a pre-trained language model rather than training a model from scratch.
- Streamlit is used to create the user interface.
- The tokenizer converts input text into model-readable tokens.
- The model generates the answer from the processed input.
- The application can be extended using a college-specific database or RAG system.